

Growing Tedana: a grassroots-leveraged platform for processing multi-echo fMRI data

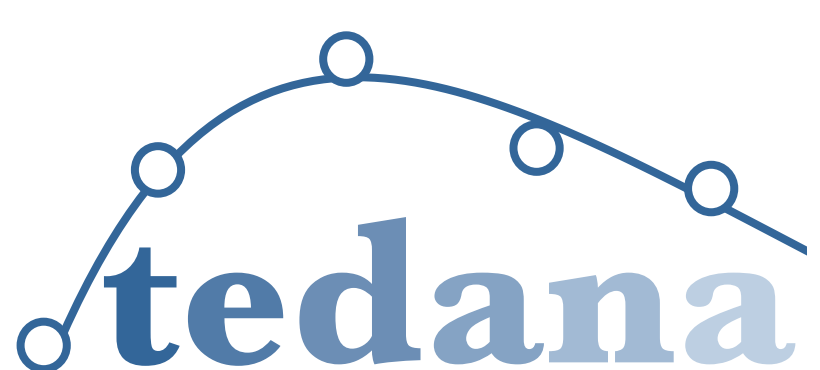
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#379

INTRODUCTION

Open software to use and improve multi-echo methods
Community & resources for multi-echo fMRI



Multi-echo fMRI, data are collected at multiple echo times (TE) after each excitation pulse. This enables time series separation of BOLD effects of potential neural interest, from non-BOLD effects¹⁻⁷. The acquisition cost of multi-echo is often minimal, and it improves data quality⁸⁻¹⁹.

Over the past year, the tedana community has made many changes that improve how researchers can use the software and understand their results. We have also made improvements increase speed and stability and support continued methodological innovation.

Additionally, we have built up resources, primarily through an interactive online book, to support the growing community of multi-echo fMRI users.

WAYS TO CONNECT

Multi-echo fMRI Book

<https://me-ica.github.io/multi-echo-data-analysis>

Multi-echo questions: <https://neurostars.org> with 'multi-echo' or 'tedana' tags

Subscribe to the tedana newsletter:

<https://groups.google.com/g/tedana-newsletter>

Join the conversation:

mattermost.brainhack.org/brainhack/channels/tedana

Recordings of multi-echo users meetings:

<https://www.youtube.com/@tedana-devs>

Openly available multi-echo fMRI datasets

<https://me-ica.github.io/open-multi-echo-data/>

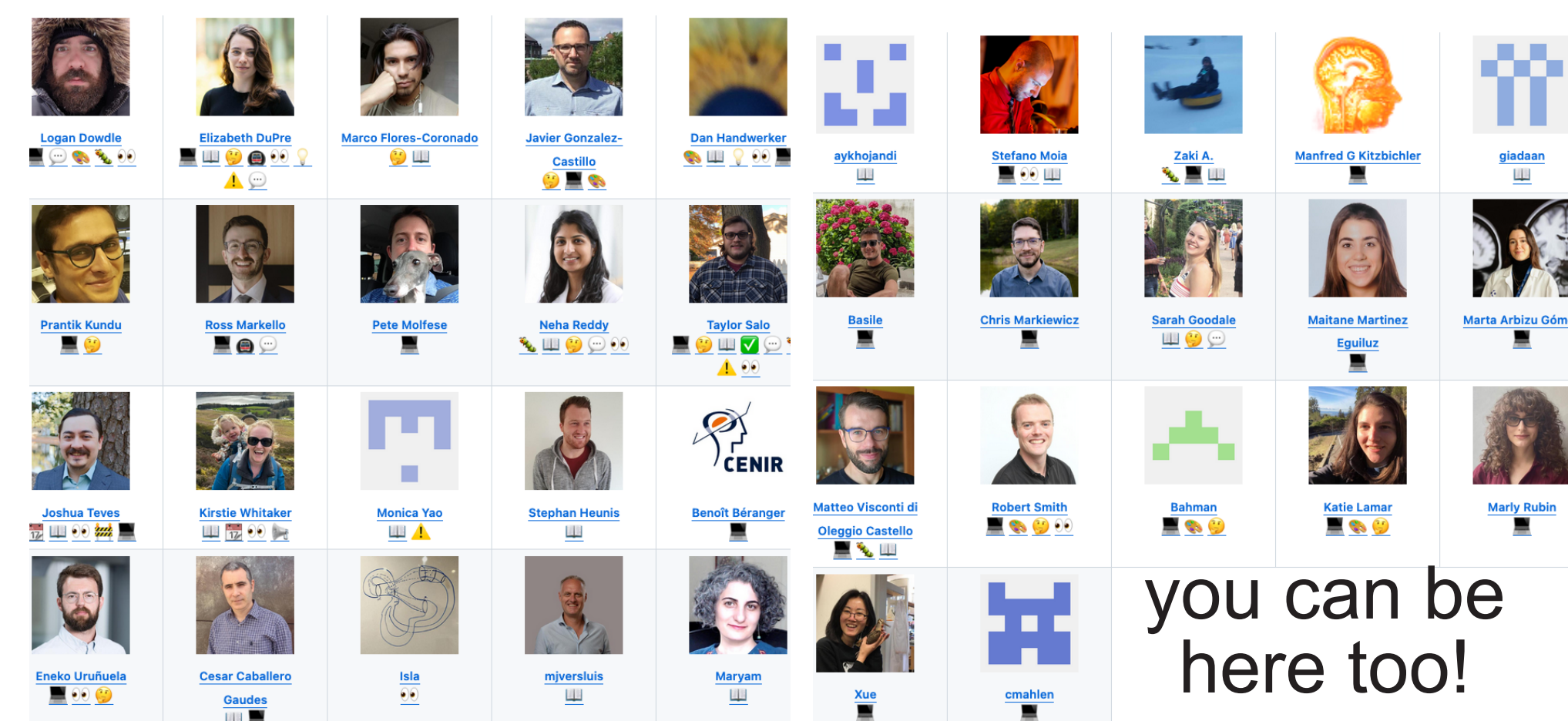
Code and resources are open source.

Contribute at: <https://github.com/ME-ICA/tedana>

Multi-echo content including tedana interactive reports, a link to this poster & a form to add info to list of multi-echo fMRI studies: me-ica.github.io/ohbm-2026-multiecho/tedana.html



CONTRIBUTORS



you can be here too!

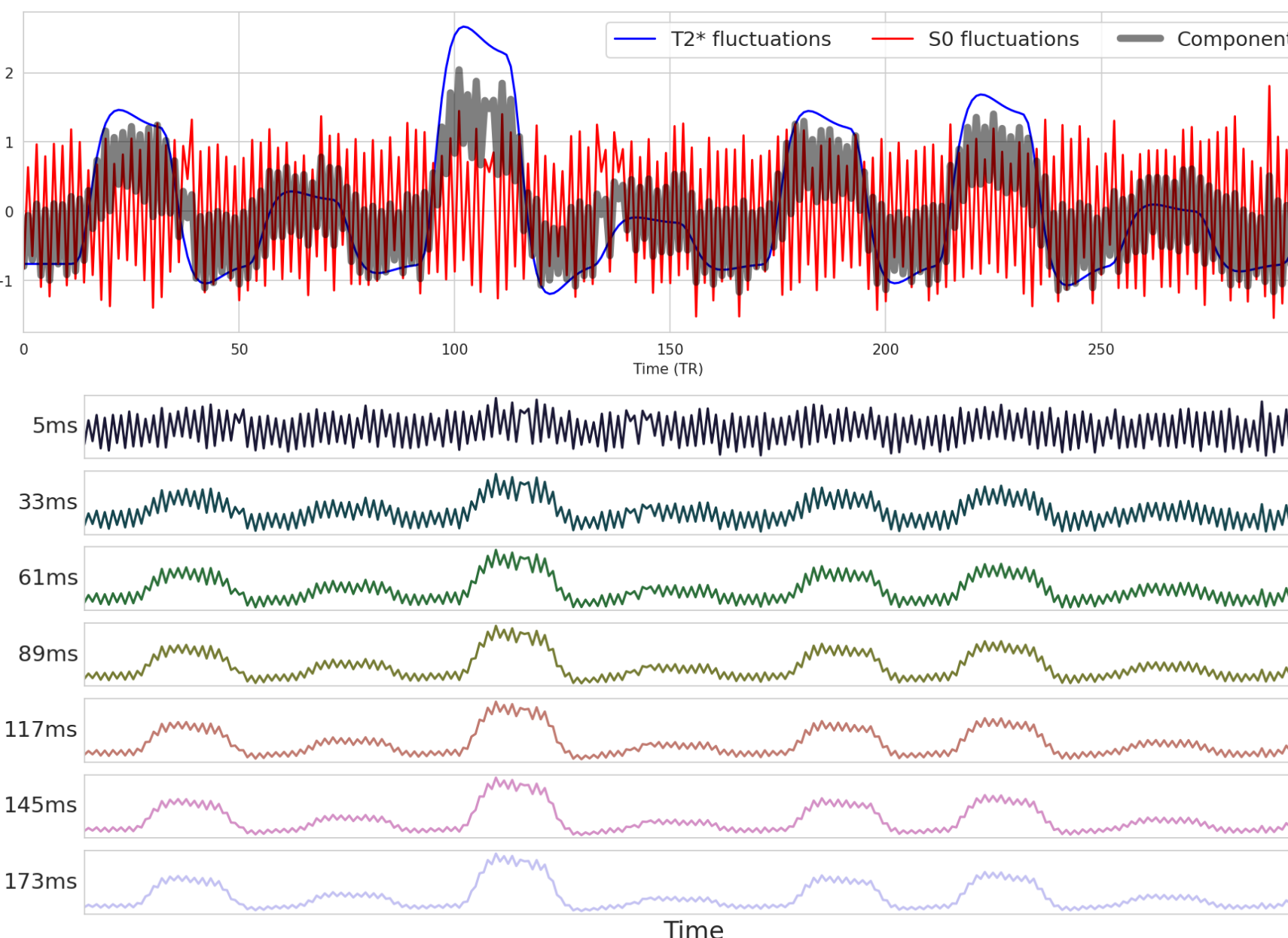
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IMPROVED MULTI-ECHO FMRI RESOURCES



Practical Resources
Open Multi-Echo Datasets
Acquiring Multi-Echo Data
Processing Multi-Echo Data
Analysis Tutorials
Optimal combination with [DuPre, 2021]
Volume-wise T2*/S0 estimation with [DuPre, 2021]
Multi-Echo Denoising with [tedana]Dual-Echo Denoising with [tedana]

Optimal combination with *t2smap*
Use [DuPre, 2021] to combine data.

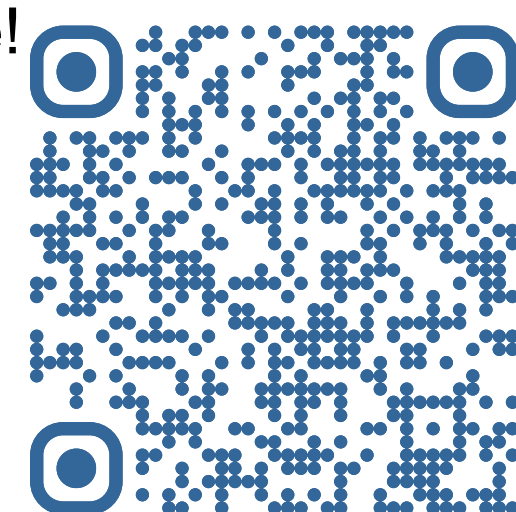
```
import json
import os
from glob import glob
import matplotlib.pyplot as plt
import nibabel as nib
import numpy as np
from book_utils import glue_figure, load_pafin
from nilearn import image, masking, plotting
from tedana import workflows
data_path = os.path.abspath('.../data')
```

Recommended Reading & History
The first two decades of multi-echo fMRI
[DuPre, 2021] includes an historical overview of multi-echo acquisition and research. Multi-echo fMRI is sometimes discussed as a newer method, but it's as old as fMRI. Much of this work set up the theory and empirical measurements that drive continued innovation in multi-echo methods. Early work trying to understand BOLD responses to neural activity used multi-echo sequences [DuPre et al., 1993]. Multi-echo fMRI is effectively extending the same frequency reduced decay curve as in fMRI spectroscopy, and some of the early work in this area called itself functional spectroscopy [DuPre et al., 1993, DuPre and Kundu, 2009].
[DuPre et al., 2009] built on much of this work to show how taking the weighted sum of multiecho fMRI improves fMRI results over single-echo fMRI. [DuPre et al., 2009] added and validated more ways to calculate the weighted sum across echoes to increase sensitivity to BOLD contrast. Power's T2* weighting method is the default weighting used in [tedana] and other software and this weighted sum is sometimes called "Optimally Combined". Consider this method being called "optimal" research continues in comparing and validating other combination options [DuPre et al., 2021].

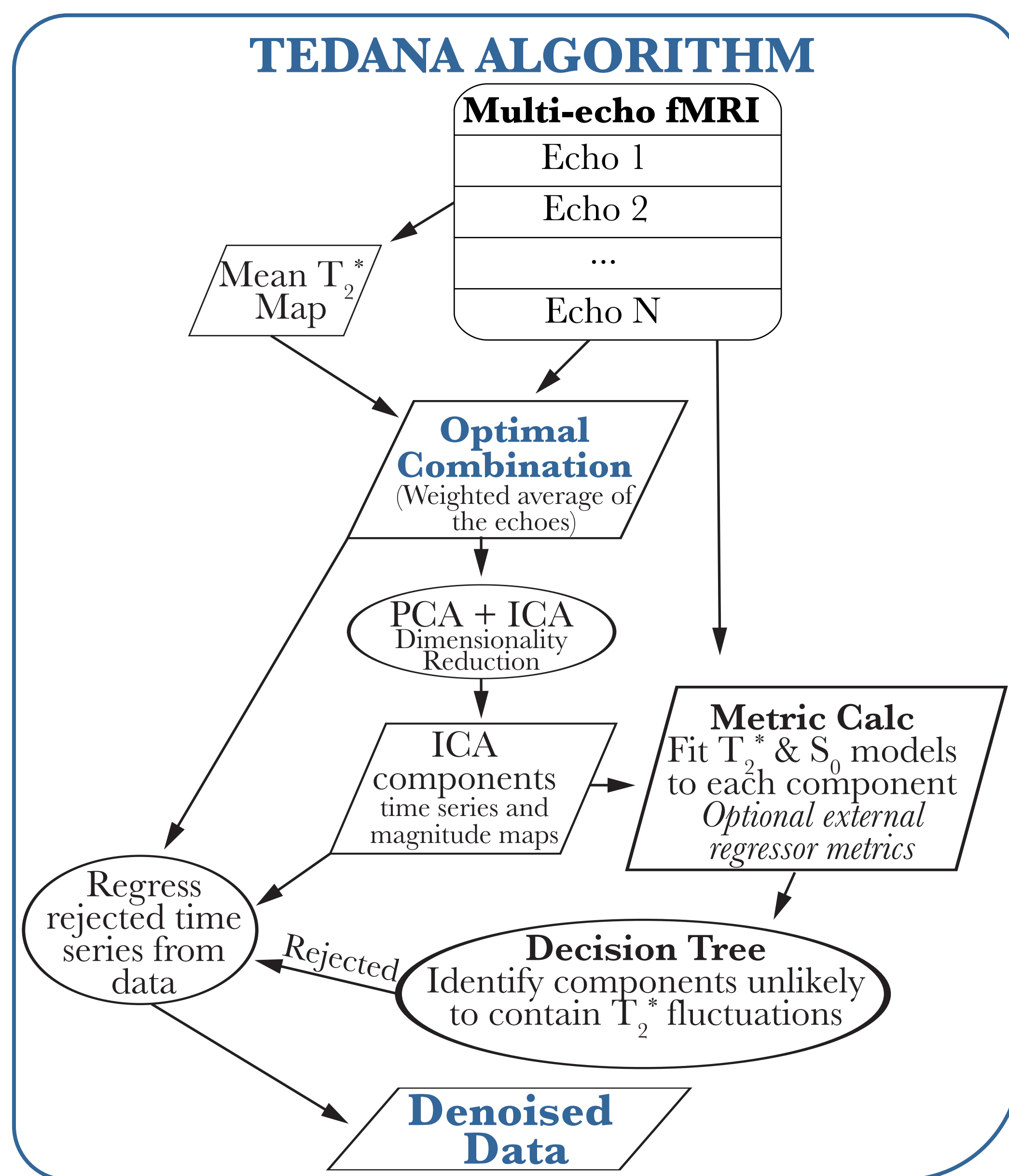
Our interactive multi-echo education book was redesigned to make it easier for more people to add content. Contributions and requests for enhancements are welcome!

Currently includes:

- An overview of MRI physics and processing underlying multi-echo fMRI
- Practical advice for acquiring multi-echo data
- Links to other multi-echo fMRI resources
- Executable pipelines for multi-echo processing methods



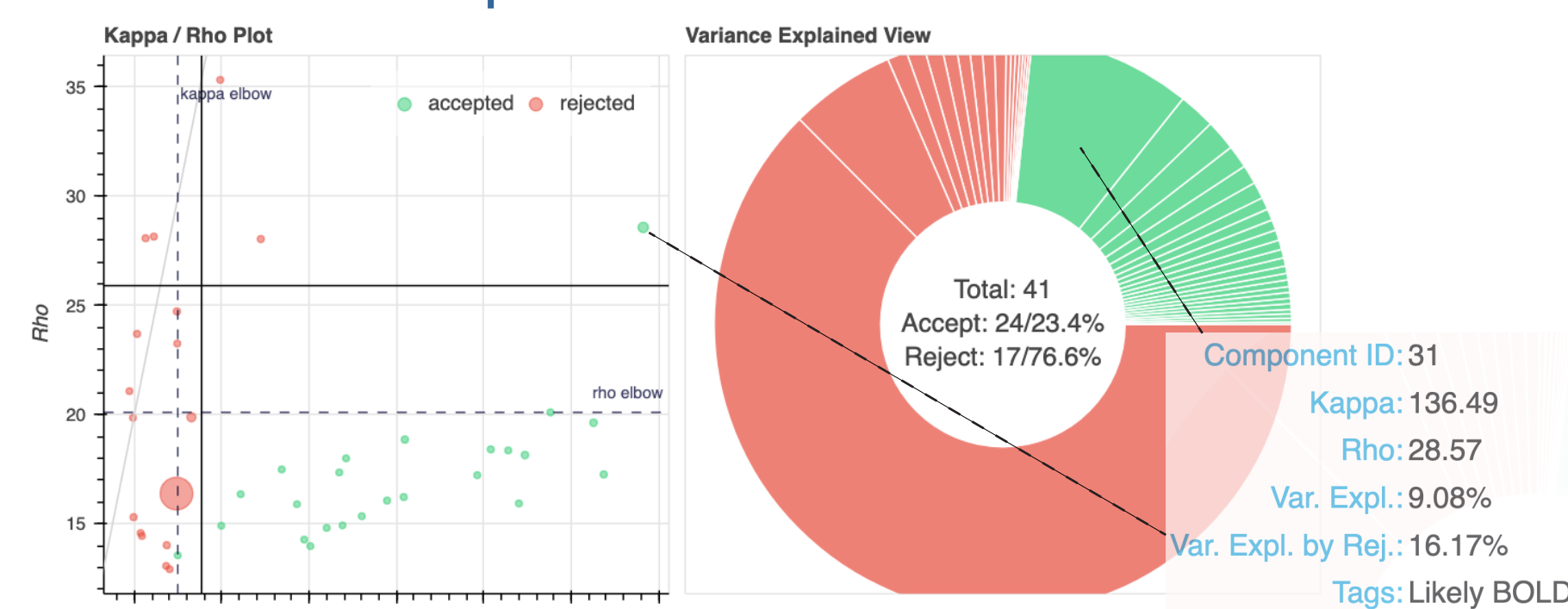
TEDANA SOFTWARE



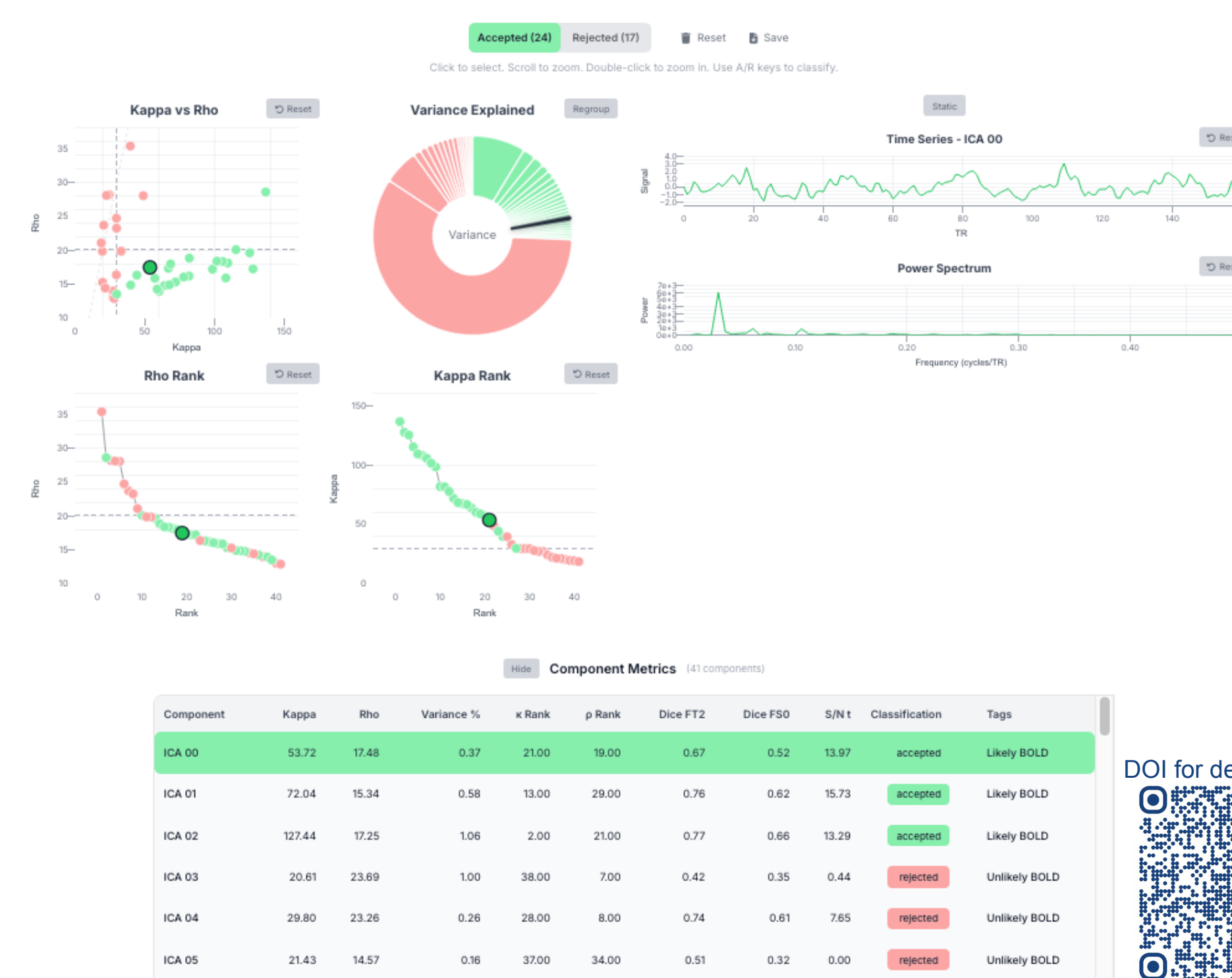
Improvements during the past year

- Faster due to better parallelization of code
- Less memory by operating only on voxels in masks
- Added dummy scan parameter to make execution easier
- Works with EPTI sequences which have many more echoes that are not independent from each other.²⁰
- User-created decision trees that identify components to reject can be added to figshare and run in tedana with just the tree name (doi.org/10.6084/m9.figshare.25251433)
- ica_reclassify can use custom tags, which makes it easier to combine tedana with other ICA denoising methods
- Switched to uv for package management, simplifying dependency management
- Changes to 96 files with 10.3K lines of code added and 2.3K lines removed

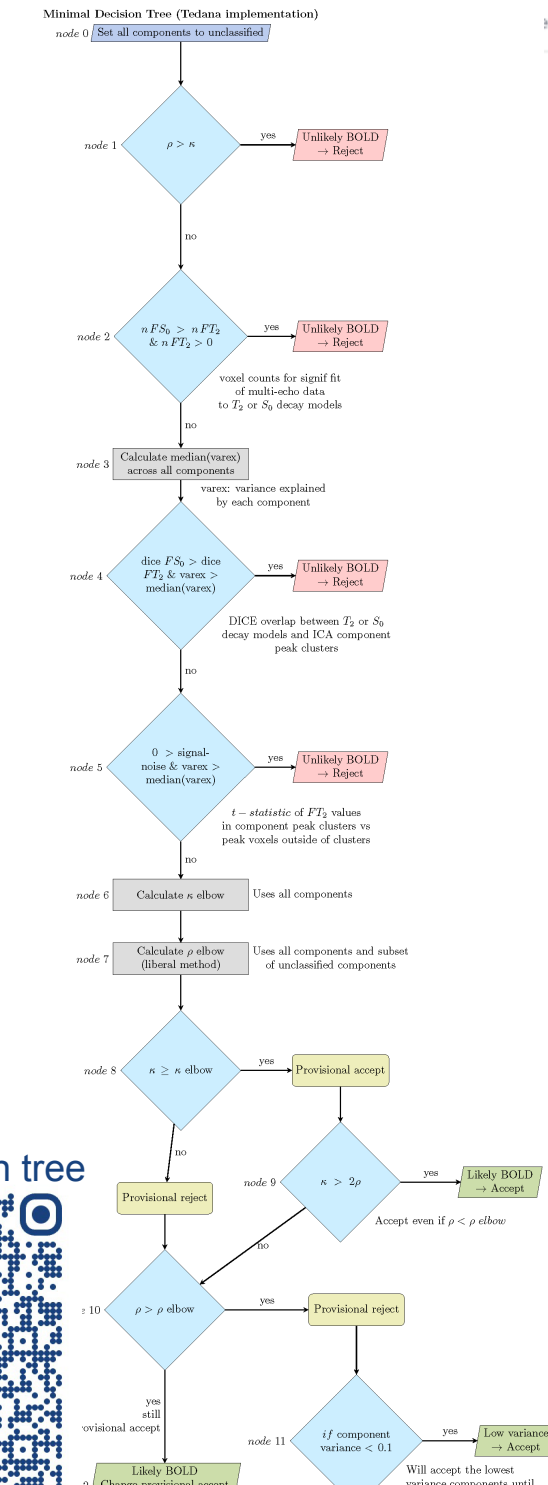
Tedana Report With Additional Information



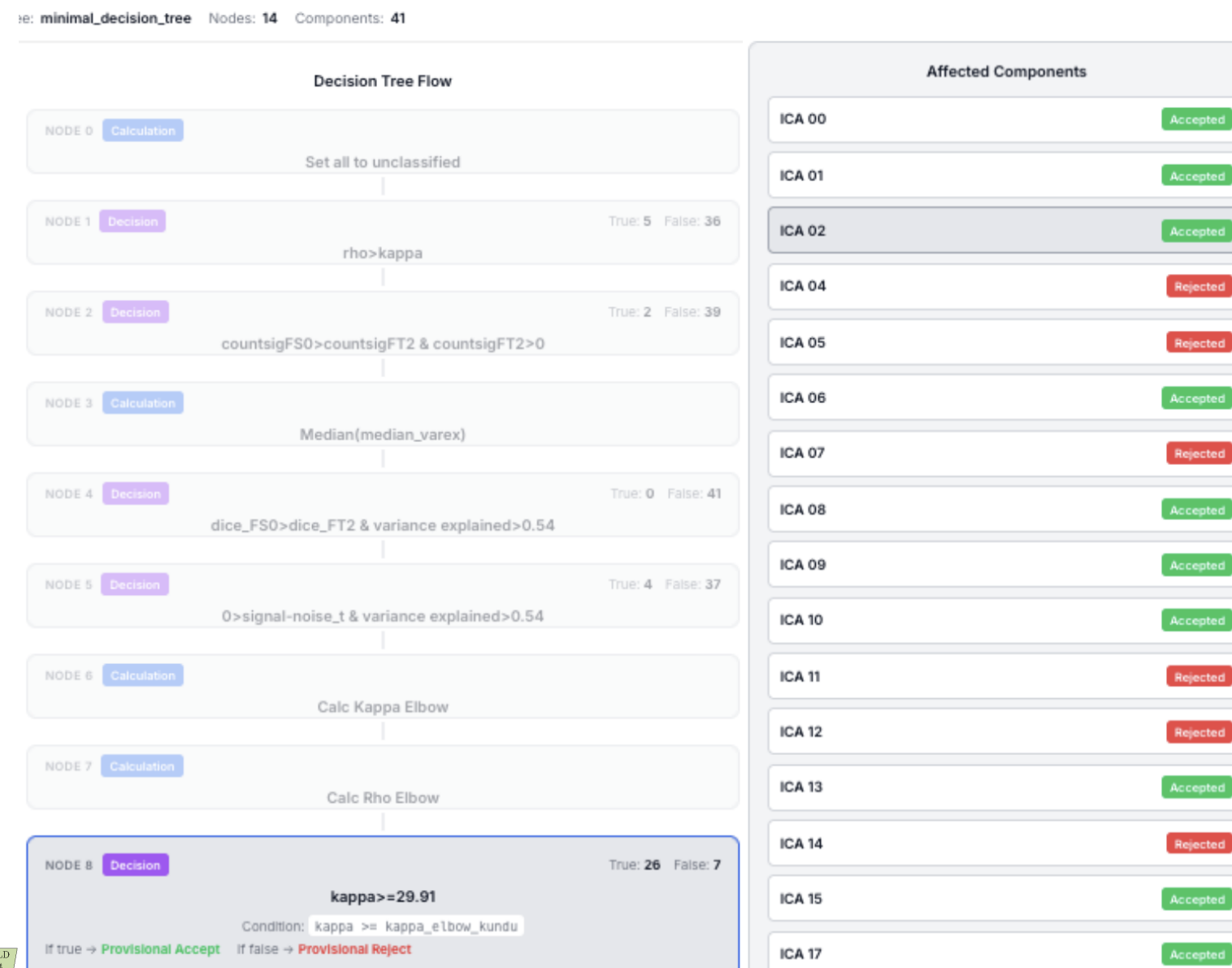
RICA Interactive Report



Decision Tree



RICA Decision Tree Report



Several improvements focused on giving users more information and easier access to information to better understand the denoising process. In tedana interactive html report, we added measures for the total variance explained by accepted & rejected components. We also added how much variance explained in each accepted component was shared by rejected components. This is useful to see how much signal is common between accepted and rejected components. Users can now use arrow keys to cycle between components in the report.

For several years, we had a separate web-server-based program, RICA that allowed people to review results and manually change component classifications. The webserver can now be run with a simple command. It also visualizes information that was stored in text files. This includes key metrics for each component and what steps in the decision tree accepted or rejected each component.

CHALLENGES & ONGOING WORK

- We want to identify components that cleanly separate BOLD from non-BOLD information. Are there better ways?
 - Improving the estimation of the number of components and how data are decomposed (poster 746)
 - Improving the component selection process (poster 644)
- What can we improve if we have better temporal resolution for echos?
 - Improving methods that use EPTI data (poster 741)
- Vascular blood oxygenation fluctuations are T2* weighted. How do we best combine respiration and cardiac measures to remove these noise sources?
- How do we quantify improved noise removal?
 - Development of a pBOLD metric¹² (poster 2481)

HOW TO CONTRIBUTE

- Share challenges & feature requests
- Request educational material
- Contribute educational material
- Contribute to the code and methods
- There is always a risk that new methods are over-tuned to specific data parameters. If you have multi-echo data, help test methods on a wider variety of real data and share what works and what can be improved.